

Exercise 2

Exercise 2.1 Simplex method

A well known method for the numerical solution of *constrained linear optimization problems* is the simplex algorithm. In the following, the linear constrained optimization problem

$$\min_{\mathbf{x} \in \mathbb{R}^4} \mathbf{c}^T \mathbf{x}, \quad \mathbf{c}^T = [1, 2, 3, 0.5] \quad (1a)$$

$$\text{s.t. } \mathbf{A}\mathbf{x} = \mathbf{b}, \quad \mathbf{A} = \begin{bmatrix} 2 & 1 & 1 & 2 \\ 2 & 0 & -3 & 2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ 1.5 \end{bmatrix} \quad (1b)$$

$$\mathbf{x} \geq \mathbf{0}, \quad (1c)$$

is to be solved with this method.

- a) Check whether each of the following combinations

$$\mathcal{B}_1 = \{1, 3\}$$

$$\mathcal{B}_2 = \{1, 4\}$$

is a basic feasible solution.

Add-On: Examine the basis $\mathcal{B}_3 = \{2, 3\}$.

- b) Solve the given optimization problem by hand using the simplex method. Start with a basic feasible solution from exercise a).

Exercise 2.2 Active set method

The following *constrained quadratic optimization problem* is given

$$\min_{\mathbf{x} \in \mathbb{R}^2} (x_1 - 2.5)^2 + (x_2 - 3.5)^2$$

$$\text{s.t. } 1) \quad 2x_1 + x_2 - 6 \leq 0$$

$$2) \quad -x_1 - x_2 + 0.5 \leq 0$$

$$3) \quad -2x_1 + x_2 - 1.5 \leq 0$$

$$4) \quad x_1 \geq 0$$

$$5) \quad x_2 \geq 0.$$

- a) Draw the admissible set into Figure 1. Is this a convex problem?
- b) Formulate the KKT conditions for the given problem.
- c) Solve the problem using the active set method with the initial point $\mathbf{x}^0 = [0.5, 0]^T$. Thereby, graphically determine the solution of the equality-constrained quadratic problem of the current working set using the contour lines and admissible set in Figure 1. Then, compute the Lagrange multipliers for the new iteration point to determine the new working set.
- d) Discuss the meaning of the Lagrange multipliers in the optimal point.

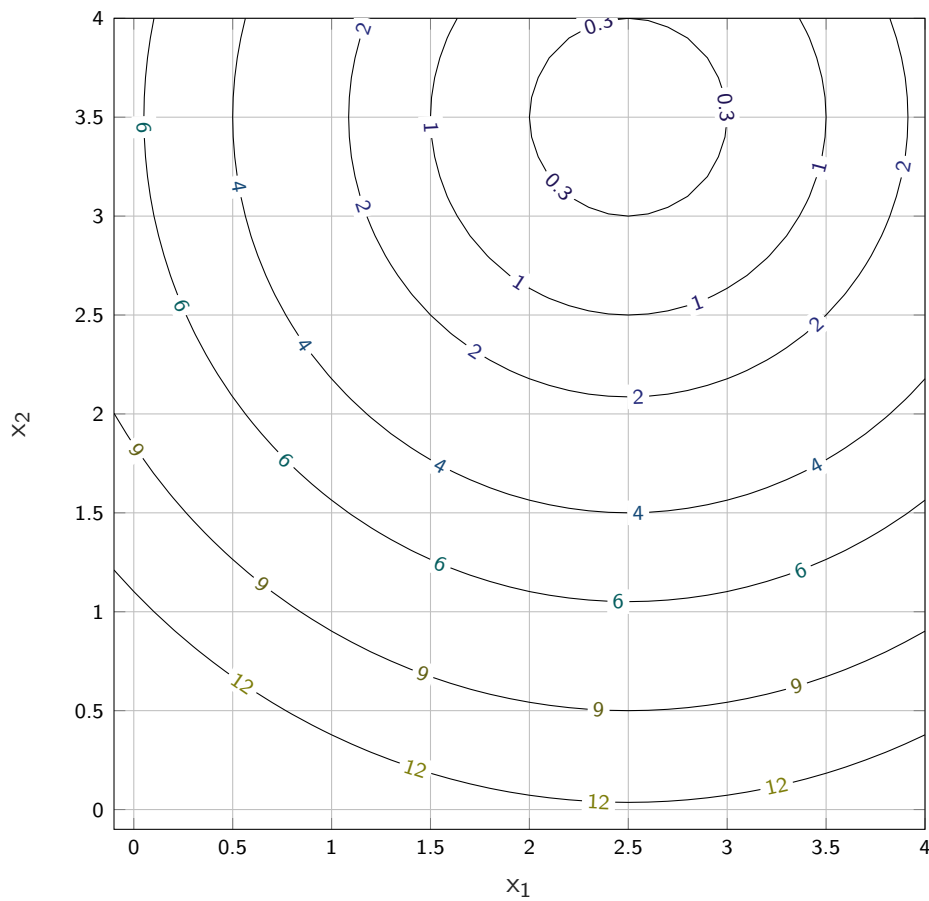


Figure 1: Contour lines of the quadratic problem from Exercise 2.2